

Practice Quiz: Planning

Part A: Multiple Choice

1. The rostro-caudal gradient of the prefrontal cortex encodes: A) Emotional intensity from low to high B) Temporal abstraction — posterior regions for concrete plans, anterior regions for abstract long-term goals C) Motor complexity from simple to compound movements D) Sensory modalities from visual to auditory
2. The basal ganglia's role in Active Inference is to: A) Store long-term memories B) Implement policy selection through direct (facilitation) and indirect (suppression) pathways C) Generate prediction errors for sensory stimuli D) Produce cerebrospinal fluid
3. Dopamine in the context of planning primarily modulates: A) Sensory acuity B) The precision of policy selection — how decisively the brain commits to a particular action plan C) Body temperature D) Language production speed
4. "Utilization behavior" in frontal lobe patients demonstrates: A) Preserved intellectual function B) Loss of hierarchical policy evaluation — objects are used automatically without contextual planning C) Enhanced motor control D) Improved creativity
5. In Active Inference, temporal discounting (preferring \$10 now over \$20 later) reflects: A) Rational economic optimization B) Low precision on long-horizon policies — the generative model's planning depth is truncated C) Superior mathematical reasoning D) Cultural values about immediacy
6. Depression-related behavioral paralysis occurs when: A) The brain generates no predictions at all B) High precision on negative outcome predictions causes all policies to appear equally futile C) The motor cortex is damaged D) The patient lacks knowledge of available options
7. Addiction progressively narrows the policy space because: A) Drug use damages all prefrontal neurons B) Dopaminergic sensitization increases precision on drug-seeking

policies relative to alternatives C) The patient forgets about other options D) Social pressure eliminates alternative behaviors

Part B: Short Answer

1. A patient with orbitofrontal cortex damage makes poor financial decisions despite normal intelligence. Analyze this in Active Inference terms: what component of policy evaluation (pragmatic value, epistemic value, or precision) is impaired?
2. Explain how behavioral activation therapy for depression (scheduling small, achievable activities) might work through Active Inference mechanisms. What precision dynamics are being modified?
3. Compare model-based planning (evaluating Expected Free Energy of future policies) with model-free habitual behavior (repeating previously reinforced actions). What neural systems support each, and when does the brain switch between them?